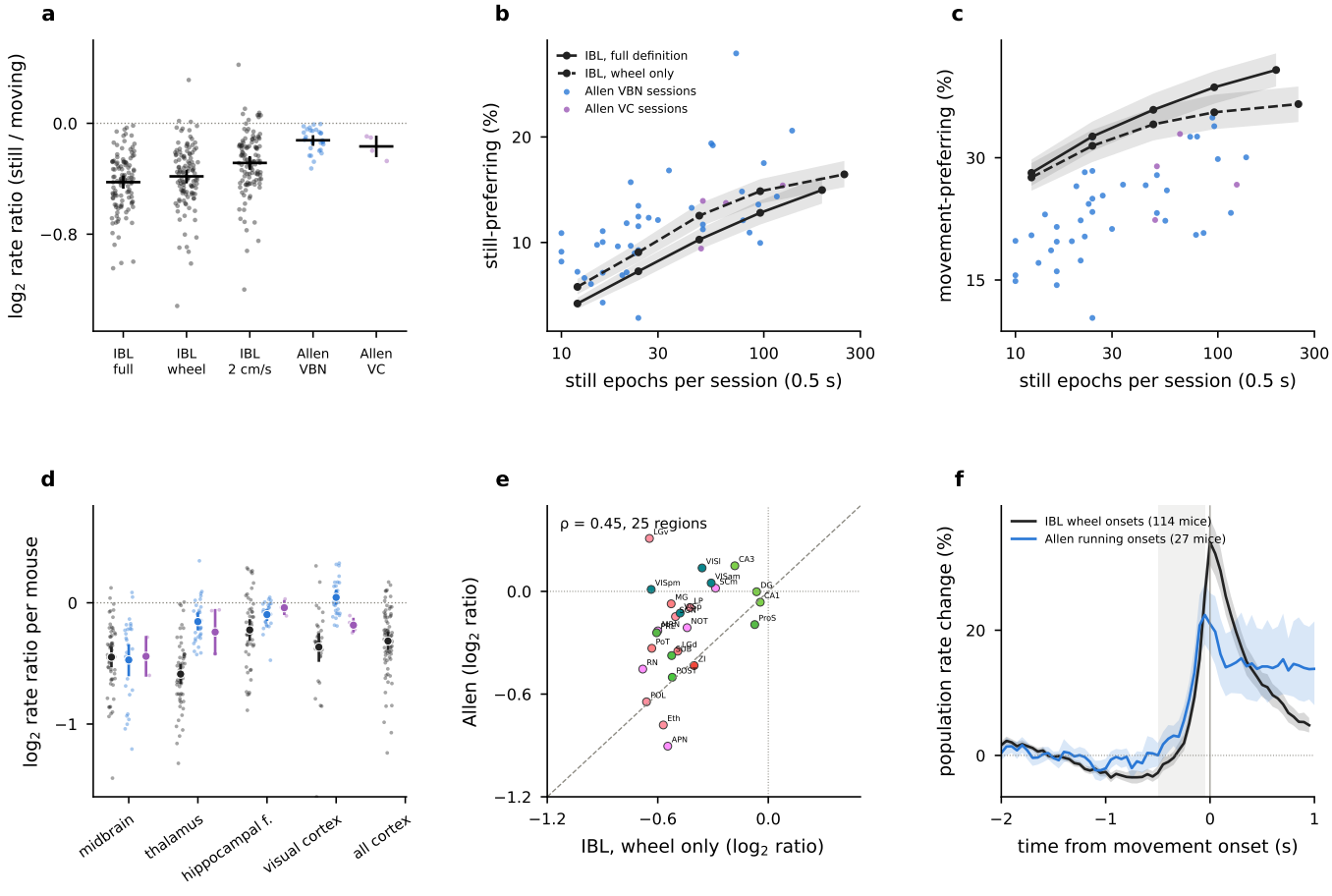


Figure 18: Neural correlates of stillness in Allen Institute Neuropixels data against IBL. **a** Mouse-level $\log_2$ rate ratio (still/moving; neurons averaged per session, sessions per mouse). IBL: the 315 sessions (115 mice) with a grey-screen wheel trace, with the report's movement definition (wheel, licks, face motion), wheel movement only ($>$0.05 rad/s), and wheel movement above the Allen running threshold (2 cm/s at the wheel rim). Allen: Visual Behavior (VBN) and Visual Coding (VC) mice, running only. Black: mean and 95% bootstrap CI over mice. **b** Still-preferring neurons (Welch $p < 0.01$, 0.5 s epochs) against the number of still epochs per session. IBL lines: still epochs randomly subsampled to 12–96, and all (session median); mean over mice and 95% CI. Dots: Allen sessions. **c** Movement-preferring neurons, as in b. **d** Rate ratio by division (sessions with $\geq$20 neurons in the division, averaged per mouse): IBL wheel only (black), Allen VBN (blue), Allen VC (purple); large dots, mean and 95% CI over mice. Allen cortical units are in visual areas; IBL is shown for visual areas and for all cortex. **e** Region-level agreement: mean over mice of the region's $\log_2$ ratio (Allen CCF labels mapped to Beryl; regions with $\geq$3 mice of $\geq$5 neurons in both datasets; Allen colours). Dashed: identity. **f** Population rate around movement onsets after $\geq$1 s of awake stillness, percent change from -2 to -1 s; mean over mice and 95% bootstrap CI. IBL onsets are wheel onsets (wheel-only definition), Allen onsets running onsets (curves from the Allen export). Shaded: -0.5 to -0.05 s test window.